

Psychometric Profiles of Senior Leaders: Patterns in Social EQ and Learning IQ

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Abstract

This report analyzes psychometric assessment data from over 60,000 senior business executives across more than 100 countries, focusing on two leadership capacities: Social EQ (interpersonal and emotional capability) and Learning IQ (cognitive adaptability and ambiguity tolerance). We address one focused question: which structural factors, specifically country of upbringing, gender, industry, and management tenure, meaningfully shape these constructs, and what do those patterns imply for finalist differentiation? Using two-way ANOVA, Welch t-tests, multivariate ANOVA, and linear regression, we find that country of upbringing is the dominant predictor, explaining roughly 10% of variance in Learning IQ (partial $\eta^2 \approx 0.10$) and approximately 3% in Social EQ (partial $\eta^2 \approx 0.027$), far exceeding the influence of gender or industry. Gender differences are statistically detectable but negligible in practical magnitude (partial $\eta^2 < 0.001$), though gaps vary meaningfully by country; Singapore shows the largest difference in both constructs. Industry affiliation is real but uniformly weak (partial $\eta^2 < 0.01$ for all traits). Social EQ and Learning IQ are stable individual traits, not products of accumulated tenure. Throughout, effect sizes are reported alongside p-values to distinguish statistically detectable from substantively meaningful differences. Findings are translated into placement-relevant recommendations for differentiating finalists and assessing mandate fit.

1 Introduction

Executive selection at the C-suite level is fundamentally a high-stakes, infrequent decision problem. By the time a search reaches its endpoint, the client is often choosing among only a handful of truly viable finalists, and the headhunter’s job is to unpack those differences and understand what separates the people at the top. In practice, psychometric assessments enter late in the funnel, typically when the client is evaluating the final 3–4 candidates, each of whom is genuinely strong, making the decision genuinely difficult. This approach values interpretation, not just measurement. The goal is to identify where these separations exist among elite profiles and understand what those differences imply in order to achieve the best fit.

This focus is consistent with the broader leadership literature. Goleman characterizes emotional intelligence as a set of competencies including self-awareness, emotion regulation, self-motivation, empathy, and relationship management, and argues that cognitive ability alone is an incomplete account of real-world success, writing that “at best, IQ contributes about 20 percent” to life success, with the remainder driven by “other forces,” including emotional competencies (1). Complementing this, trait-based research situates leadership differences within stable personality dimensions; Judge et al. (2) synthesize evidence using the Big Five framework and report systematic associations between personality traits and leadership outcomes. Many executive assessments are repackaged versions of well-known trait models, so interpreting results in that established framework improves clarity and credibility. This has informed the choice to focus in on two measures: Social EQ and Learning IQ.

Against this background, Dr. Dana Landis framed the expected output for this project as pattern discovery plus interpretation: use available “divides” (e.g., region, gender, and levels) to identify “relationships and patterns” in the scores and then ask “what does it mean?” and “what story is that telling us?”

Accordingly, this report addresses one focused question: *which structural factors meaningfully shape Social EQ and Learning IQ among senior executives, and what do those patterns imply for finalist differentiation?* With a focus on country of upbringing, gender, industry, and management tenure, the analysis aims to answer these questions. We quantify effect sizes alongside p-values to separate what is statistically detectable from what is practically meaningful. The goal was to translate the most robust findings into context-aware interpretations designed to help placement professionals differentiate finalists and assess best fit.

2 Data

2.1 Sources and Sample

Two datasets were used. The base dataset (`Leadership_141XP`) contains 60,217 valid records after removing test administrations (`isTest = TRUE`). After only retaining assessments with complete scores and known binary gender, the dataset contains 57,257 complete case observations. The augmented dataset (`Leadership_141XP_Augmented_01`) contains 39,235 records with extended 23-dimension profiles; all analyses requiring additional personality, motivation, and competency scales draw on this subset. Both files use a space-delimited format with double-quote text qualification. The executive cohort has a mean age of approximately 56, a median of 11 years in management, and a notable inter-

national footprint: 9.1% were raised outside their citizenship country and 17.3% currently work abroad. Full sample characteristics are provided in Appendix B (Tables 7 and 8).

2.2 Instruments and Scales

The assessment produces two composite super-factor scores that are the primary focus of this report:

- **Social EQ** – interpersonal and emotional capability composite
- **Learning IQ** – cognitive adaptability and ambiguity tolerance composite

The dataset additionally provides Strives scales (motivational orientation, 6 dimensions), and the augmented version includes PersonalityCore scales (6 dimensions), and Competency scales (9 dimensions across two instruments). All scores are standardized relative to a broader reference population: a score of 0 represents the reference baseline, or average, and positive values indicate above-average standing, while negative values indicate below-average standing.

Note on sample size and significance: With over 56,000 observations, even effects explaining less than 0.1% of variance will achieve $p < 0.001$. Every p-value in this report is paired with a practical effect size (partial η^2) to distinguish *statistically detectable* from *practically meaningful* differences.

3 The Executive Psychometric Profile

All psychometric scores are standardized against a broader reference population, so the mean scores of this sample directly indicate how far above or below average senior executives fall on each trait.

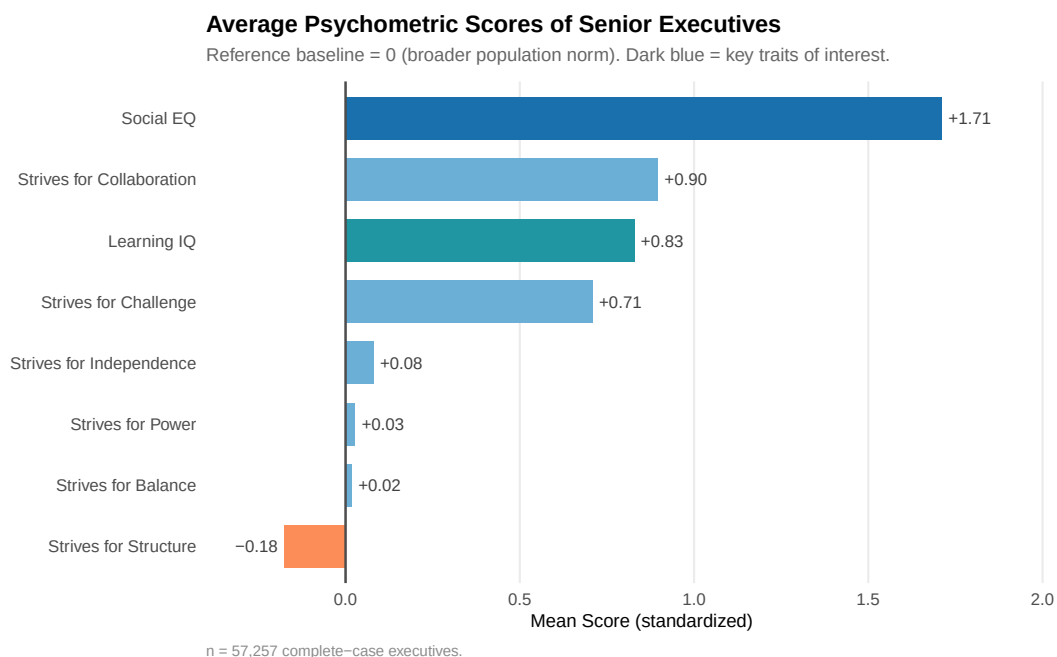
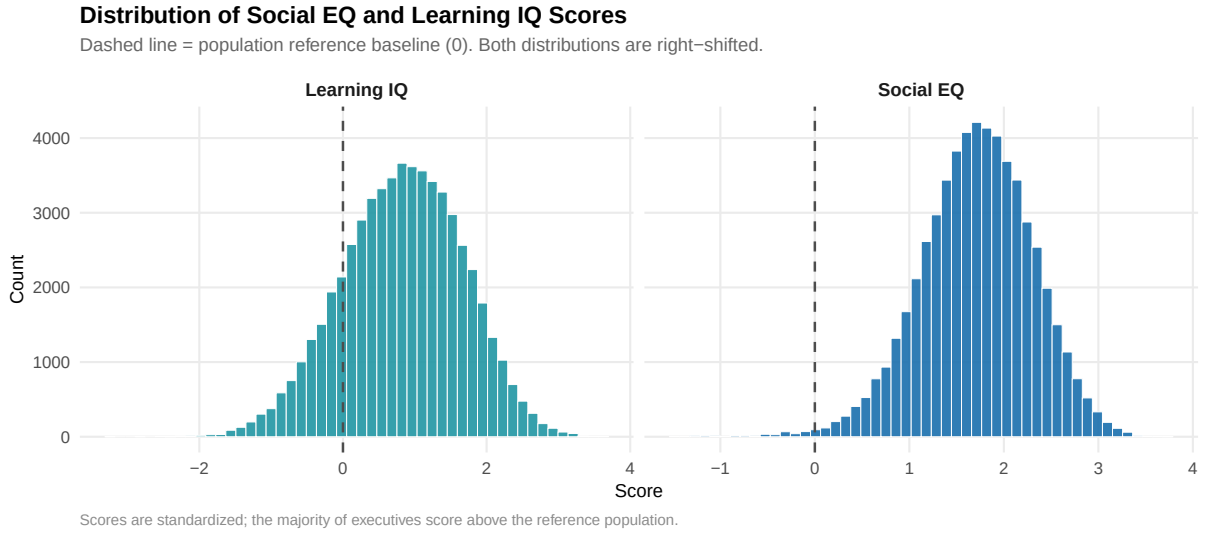


Figure 1: Average Psychometric Scores of Senior Executives

Social EQ (mean ≈ 1.71) stands out as the trait where executives most dramatically outperform the reference baseline, roughly 1.7 standard deviations above average. The executives in this dataset are an unusually emotionally intelligent group by population standards. Learning IQ (mean ≈ 0.83) is also meaningfully above baseline, indicating well above-average cognitive adaptability and openness to learning. Among motivational traits, Strives for Collaboration and Strives for Challenge are most elevated, suggesting executives are oriented toward teamwork and growth. Strives for Structure is the only trait below the reference baseline, consistent with the demands of senior leadership. Score distributions confirm the rightward shift of both primary constructs; demographic characteristics are summarized in Appendix B (Figures 2 and 4).

Figure 2: Distribution of Social EQ and Learning IQ Scores



Taken together, these results suggest that executives exhibit a distinct profile marked by strong interpersonal effectiveness, learning agility, and growth-oriented motivation. The elevated Social EQ pattern highlights the importance of emotional awareness and relationship management in executive roles, while the higher Learning IQ reflects adaptability and quick learning in complex and changing environments. In contrast, the lower tendency toward structure suggests comfort operating under ambiguity and strategic flexibility. With this baseline established, the central question is whether these profiles vary systematically across the structural divides available in the data.

4 Gender and Country of Upbringing

4.1 Overall Gender Differences

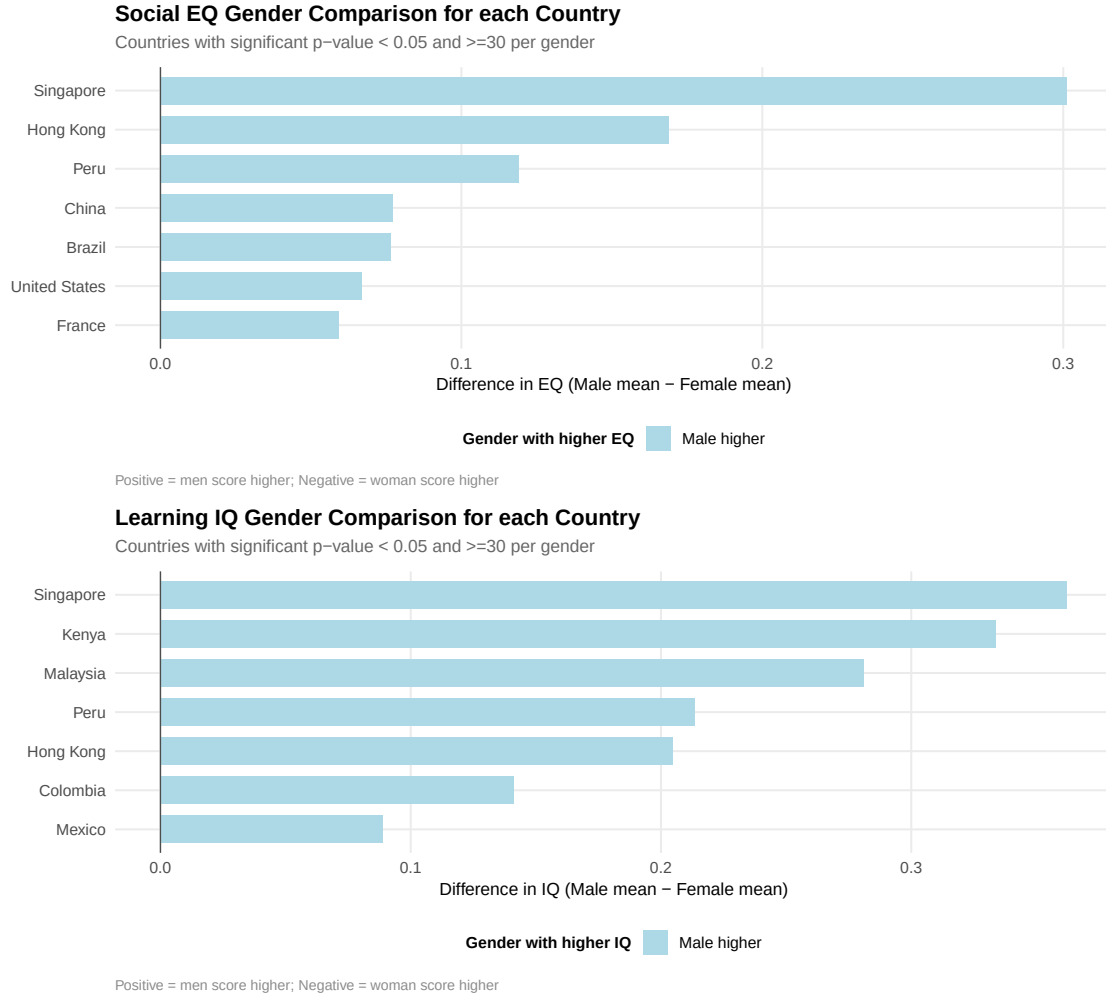


Figure 3: Social EQ and Learning IQ Gender Comparison for each Country

Gender is statistically significant for both outcomes, but the partial η^2 for gender is less than 0.001 for both Social EQ and Learning IQ — a negligible practical effect. The distributions of scores overlap almost entirely between male and female executives. This is a characteristic large-sample artifact: with 59,000+ observations, even a difference of a fraction of a point registers as highly significant. The box plots visualizes this; medians are nearly identical and the interquartile ranges overlap substantially. Full ANOVA results are reproduced in Appendix B (Table 9).

4.2 Country of Upbringing: The Dominant Factor

Country of upbringing is a far more consequential predictor. For Social EQ, country explains approximately 2.7% of variance (partial $\eta^2 \approx 0.027$, a small but real effect). For Learning IQ, country explains approximately 9.6% of variance (partial $\eta^2 \approx 0.096$, a **moderate** effect by conventional benchmarks).

This is a substantive structural finding: where an executive was raised shapes their Learning IQ profile more than any other single factor measured. The magnitude is several times greater than the gender effect and approaches the threshold of practical significance in a domain where individual trait scores are used to differentiate already-elite candidates. For a headhunting firm, this implies that geographic diversity in candidate sourcing may yield meaningfully different cognitive and learning profiles. A candidate’s country of upbringing is a useful contextual lens when interpreting their Learning IQ score.

4.3 Gender Gaps Across Countries

While the overall gender effect is small, the *pattern* of differences varies significantly by country (Country $\times$ Gender interaction, $p < 0.0001$ for both outcomes). The chart below shows the 15 countries with the largest absolute gender gaps, restricted to countries with at least 30 male and 30 female executives assessed.

Singapore shows the largest gender gap in both Social EQ and Learning IQ, with male executives scoring higher. The direction of the gap is not uniform across countries: in several countries female executives score higher on Social EQ, underscoring that the pattern is country-specific rather than a global trend. Welch t-tests identifying countries with statistically significant gender gaps are provided in Appendix A (Tables 2 and 3).

Industry affiliation provides a second structural lens, though as shown next, has a modest explanatory power by comparison.

5 Industry and Organizational Context

A MANOVA tested whether executives’ full psychometric profiles differ by employer industry. The result is highly significant (Pillai’s Trace = 0.046, approx. $F(136, 456, 480) = 19.43$, $p < 2.2 \times 10^{-16}$), confirming that industry context is systematically associated with the psychometric characteristics of the executives it attracts and develops. All traits differ significantly across industries, but the effect sizes are uniformly small (partial $\eta^2 < 0.01$ for all). Within-industry individual variation is substantial, and industry-level norms should be treated as weak prior information rather than strong predictors. Detailed MANOVA results and ANOVA results on the industry effect of executive psychometric traits are provided in Appendix A (Table 4, Table 5).

Executives in Telecommunications and Consumer Goods score highest on Social EQ, over one and a half standard deviations above the reference mean. Advanced Technology, and other professional service industries like Healthcare and Financial rank leading on Learning IQ. These patterns are directionally interpretable: industries requiring intensive relationship management skew toward higher Social EQ; industries with rapidly changing knowledge demands skew toward higher Learning IQ. Detailed ANOVA results and industry-level mean charts are provided in Appendix A and Appendix B (Table 5, Figure 6).

A third question remains: does tenure in management predict these scores, or are they fixed characteristics of the individual?

6 Management Experience and Psychometric Scores

A natural question for a placement firm is whether executives develop higher Social EQ and Learning IQ with more years of management experience. The data give a clear answer: the relationship is statistically significant, but $R^2 \approx 0.005$ for both traits, meaning that experience explains less than 1% of the variance in either score. The trend lines are essentially flat. This is a null result, but rather meaningful, suggesting that Social EQ and Learning IQ are not competencies that accumulate primarily through tenure and appear largely independent of years of management experience. Years of experience is not a reliable measure for either capacity, and differences in seniority among finalists should not be interpreted as differences in these two constructs.

For the client, this has a direct implication: when two finalists differ on Social EQ or Learning IQ, that difference likely reflects an individual characteristic rather than a career-stage artifact. As experience is not meaningfully related to either score, differences in seniority should not be interpreted as evidence that lower scores will converge over time. Hiring decisions should therefore evaluate these traits directly rather than relying on years of executive leadership experience as a proxy, reinforcing a more meritocratic model of hiring. Regression results are provided in Appendix A and Appendix B (Table 6, Figure 7).

7 Discussion and Conclusion

7.1 Key Findings

Three findings from this analysis have direct implications for executive search and finalist differentiation.

Country of upbringing is the strongest contextual lens. It is the single strongest predictor of Learning IQ in this dataset (partial $\eta^2 \approx 0.096$). When comparing finalists from different geographic backgrounds, their Learning IQ scores should be interpreted with awareness of this structural context — a score at the 70th percentile for a candidate raised in one country may represent a different relative standing than the same score for a candidate raised elsewhere. This argues for benchmarking within relevant subgroups where sample sizes permit.

Social EQ and Learning IQ are stable trait-level differences, not experience accumulations. The near-zero relationship with years in management ($R^2 \approx 0.005$) means that when two finalists differ on these dimensions, that difference is more likely to reflect enduring individual characteristics than career-stage artifacts. This makes the constructs more useful as differentiators, but also means the gap is unlikely to close over time.

Industry context shapes profiles modestly. Executives self-select and are selected into industries, and this shows in the data. When evaluating a candidate’s fit for a mandate that differs from their industry of origin, industry-level norms provide useful interpretive context — but industry alone cannot substitute for individual assessment.

Taken together, the analysis supports the client’s framing: the value of psychometric data at the finalist stage lies in identifying where separations exist and what those separations plausibly mean for mandate fit. The constructs are well-suited to that purpose — they are stable, they differentiate at the elite end of the distribution, and they connect to interpretable dimensions of leadership behavior.

7.2 Advice

- (i) **Benchmark to peer groups, not population averages.** Translate raw scores into percentile ranks within the most relevant peer group. The actionable debrief output is “top decile on Learning IQ relative to peers with a similar upbringing context,” not “score = 0.83.” Pair each score with the two or three Strives deviations that most distinguish the candidate from the other finalists.
- (ii) **Interpret scores through the lens of country of upbringing.** Country of upbringing is the single strongest structural predictor in the dataset — it explains approximately 10% of variance in Learning IQ (partial $\eta^2 \approx 0.096$) and approximately 3% in Social EQ (partial $\eta^2 \approx 0.027$), dwarfing all other factors. When finalists come from different geographic backgrounds, run both a within-country benchmark and a global benchmark, and report both when they diverge. A given Learning IQ score means something different depending on the upbringing context.
- (iii) **Use gender and industry as sourcing context, not evaluation criteria.** The overall gender effect is negligible (partial $\eta^2 < 0.001$); gender should not drive shortlist decisions, though country-specific gaps (e.g., Singapore) can be noted as exploratory context. Industry effects are real but uniformly small ($\eta^2 < 0.01$ for all traits) — more useful for upstream sourcing (which sectors attract higher Learning IQ profiles) than for differentiating finalists already in the room.
- (iv) **Do not use years of experience as a proxy for Social EQ or Learning IQ.** Management experience explains less than 1% of variance in both constructs ($R^2 \approx 0.005$). When two finalists differ on Social EQ or Learning IQ, that gap reflects a stable individual characteristic, not a career-stage artifact. Seniority differences should not be interpreted as evidence that lower scores will converge over time.

7.3 Conclusion

This report set out to answer one question for the client: which structural factors meaningfully shape Social EQ and Learning IQ among senior executives, and what do those patterns imply for finalist differentiation?

Four structural factors were examined, capturing different information. Country of upbringing is the dominant predictor by a wide margin, accounting for roughly 10% of variance in Learning IQ (partial $\eta^2 \approx 0.10$) and approximately 3% in Social EQ (partial $\eta^2 \approx 0.027$), a moderate effect in a domain where individual differences drive placement decisions. This aligns with cross-cultural research showing that the environment in which one is raised shapes fundamental patterns of cognitive orientation and adaptability (3). Gender is globally negligible in practical magnitude (partial $\eta^2 < 0.001$), though the more actionable insight is that gender gaps vary meaningfully by country. Singapore shows the largest difference in both constructs, meaning global averages overshadow local heterogeneity. Industry affiliation is real but uniformly weak (partial $\eta^2 < 0.01$ for all traits), reflecting the self-selection and selection dynamics of how executives end up in sectors rather than causal shaping of their traits. Management tenure does not have an association with Social EQ and Learning IQ: years of experience accounts for less than 1% of variance in either construct ($R^2 \approx 0.005$), consistent with longitudinal evidence that trait-level individual characteristics maintain strong rank-order stability across adult career stages (4).

What does this mean for the client? The executives in this dataset are already elite: Social EQ averages approximately 1.71 standard deviations above the broader reference population, and Learning IQ averages approximately 0.83 standard deviations above it (Figure 1). Differentiation at that level is inherently demanding, and the data show that meaningful signals exist, but only if the right interpretive lens is applied. The actionable signal lives in (i) where a candidate stands relative to peers with the same upbringing context, (ii) how their motivational Strives or personality profile deviates from other finalists in the shortlist and whether those attributes best align with company culture and dynamics, and (iii) what the country-specific gender context says about comparability. Industry background and years of experience, by contrast, contribute little additional information once individual scores are known.

The overall implication is that the assessment toolkit is well-suited to finalist differentiation and that the constructs are relatively stable. The structural factors analyzed here provide a principled framework for contextualizing scores rather than reading them in isolation. Translating that framework into placement decisions requires pairing the quantitative findings above with case-to-case judgment: the data tell you where the separations are; the client's criteria determine which separations matter.

References

- [1] Goleman, D. (2020), *Emotional Intelligence: Why It Can Matter More Than IQ (25th anniversary edition)*, Bantam.
- [2] Judge, T. A., Bono, J. E., Ilies, R. and Gerhardt, M. W. (2002), ‘Personality and leadership: A qualitative and quantitative review’, *Journal of Applied Psychology* **87**, 765–780.
- [3] Nisbett, R. E. (2003), *The Geography of Thought: How Asians and Westerners Think Differently...and Why*, Free Press.
- [4] Roberts, B. W. and DelVecchio, W. F. (2000), ‘The rank-order consistency of personality traits from childhood to old age: A quantitative review of longitudinal studies’, *Psychological Bulletin* **126**(1), 3–25.

A Statistical Methods

Table 1: Statistical Methods Summary

Analysis	Method	Software
Gender $\times$ Country effects	Two-way ANOVA with interaction	Base R <code>aov()</code>
Country-level gender differences	Welch two-sample t-test per country	Base R <code>t.test()</code> via <code>broom::tidy()</code>
Industry psychometric effects	One-way MANOVA, then univariate ANOVA	Base R <code>manova()</code> , <code>anova(lm())</code>
Management experience	Ordinary least squares regression	Base R <code>lm()</code>
Effect size	Partial $\eta^2 = (df \times F) / (df \times F + df_{\text{residual}})$	Computed in R

Sample: All analyses use complete-case observations (listwise deletion of rows with any missing psychometric score). Test respondents (`isTest == TRUE`) were excluded prior to all analyses.

On p-values and sample size: With $n \approx 56,000$, the statistical power to detect arbitrarily small effects is extremely high. A p-value alone does not indicate practical importance. Partial η^2 and R^2 are the appropriate guides to substantive significance throughout this report.

Table 2: Welch Two Sample t-test on Social EQ and gender per country

Country	Citizenship	estimate	estimate 1	estimate 2	statistic	p-value	parameter	conf low	conf high
	Brazil	-0.07662850	1.553944	1.630572	-3.300711	9.936556e-04	1163.576578	-0.12217792	-0.031079077
	China	-0.07721897	1.505003	1.582222	-2.132356	3.332578e-02	698.481568	-0.14831832	-0.006119624
	France	-0.05919225	1.604789	1.663982	-2.453467	1.429264e-02	1178.602247	-0.10652688	-0.011857616
	Hong Kong	-0.16901386	1.436228	1.605241	-2.299857	2.235676e-02	229.017706	-0.31381456	-0.024213158
	Israel	-0.53892143	1.449479	1.988400	-2.519221	1.972856e-02	21.491979	-0.98318113	-0.094661725
	Lithuania	-1.75772500	0.535425	2.293150	-3.512405	4.890576e-02	2.600357	-3.49816127	-0.017288727
	Peru	-0.11912501	1.718276	1.837401	-2.160291	3.179563e-02	227.386030	-0.22778168	-0.010468332
	Singapore	-0.30135429	1.302770	1.604124	-3.753452	2.407060e-04	166.359233	-0.45986748	-0.142841093
	United States	-0.06689553	1.756728	1.823624	-5.460108	4.974464e-08	5316.960479	-0.09091386	-0.042877198

Table 3: Welch Two Sample t-test on Learning IQ and gender per country

Country	Citizenship	estimate	estimate 1	estimate 2	statistic	p-value	parameter	conf low	conf high
	Colombia	-0.14106912	0.5231633	0.6642325	-2.808240	0.0051416774	606.385020	-0.2397228	-0.042415394
	Guatemala	-0.88257500	-0.1900250	0.6925500	-2.760142	0.0193521711	10.463711	-1.5907816	-0.174368388
	Hong Kong	-0.20486588	0.2232302	0.4280961	-1.982075	0.0485901892	244.586047	-0.4084538	-0.001277985
	Kenya	-0.33379532	0.1993958	0.5331911	-3.416834	0.0008022373	160.968500	-0.5267174	-0.140873266
	Malaysia	-0.28100975	0.1339050	0.4149148	-3.639003	0.0003192472	318.246712	-0.4329391	-0.129080368
	Mexico	-0.08884668	0.4298316	0.5186783	-2.829618	0.0047389790	1185.211507	-0.1504502	-0.027243202
	Peru	-0.21367791	0.3769532	0.5906311	-3.005377	0.0029706408	213.056979	-0.3538246	-0.073531238
	Singapore	-0.36237968	0.3299089	0.6922886	-3.212307	0.0015727869	171.482688	-0.5850543	-0.139705061
	United Arab Emirates	0.83082241	1.4186500	0.5878276	-4.250620	0.0059238434	5.747412	0.3474096	1.314235274

Table 4: MANOVA

MANOVA: Industry Effect on Executive Psychometric Profiles						
Pillai's trace; predictor = Employer Industry						
Term	Pillai's Trace	approx F	num Df	den Df	p-value	
employerIndustry	0.046	19.43	136	456480	2.2e-16	

Table 5: Industry Effect on Executive Psychometric Traits

Industry Effect on Executive Psychometric Traits			
One-way ANOVA per trait; predictor = Employer Industry (18 levels)			
Trait	F(17)	p-value	Partial η^2 ¹
Strives for Collaboration	35.7	2.77e-117	0.0105
Strives for Challenge	31.2	2.20e-101	0.0092
Strives for Power	22.5	1.40e-70	0.0067
Strives for Independence	19.4	1.05e-59	0.0058
Learning IQ	17.6	1.66e-53	0.0052
Strives for Balance	12.9	4.59e-37	0.0038
Strives for Structure	12.3	3.37e-35	0.0037
Social EQ	5.8	1.41e-13	0.0017
¹ Partial η^2 reflects practical effect size: < 0.01 negligible, 0.01–0.06 small.			

Table 6: OLS Regression on Score on Years in Management

OLS Regression: Score $\sim$ Years in Management					
Coefficient on years in management, with model fit					
Outcome	β (slope)	Std. Error	t-statistic	R^2 ¹	p-value (model)
Social EQ	0.0097	3e-04	28.85	0.0144	1.04e-181
Learning IQ	0.0129	5e-04	27.00	0.0126	1.45e-159
¹ $R^2 < 0.01$ indicates experience explains less than 1% of variance in either score.					

B Supplementary Figures and Tables

Table 7: Dataset at a Glance

Characteristic	Value
Total assessments (raw)	60,217
Complete-case observations	57,257
Countries represented	153
Industries represented	18
Male executives	43,733
Female executives	12,626

Table 8: Executive Demographic Highlights

Characteristic	Value
Mean age	55.8 years
Median years in management	11 years
Raised in a different country (<code>movedCountry</code>)	5,188 (9.1%)
Currently works abroad (<code>worksAbroad</code>)	9,880 (17.3%)

Figure 4: Age Distribution (Left) and International Mobility Profile (Right)

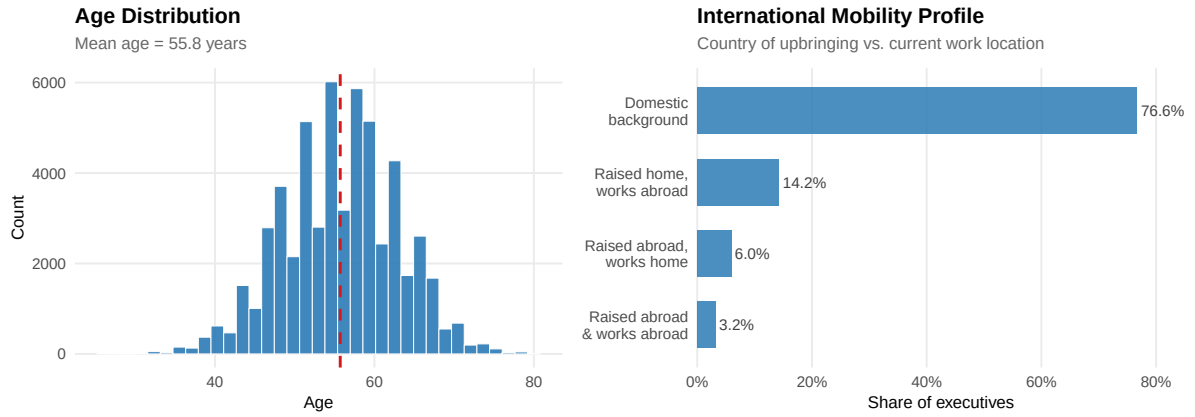


Table 9: Two-way ANOVA

Two-Way ANOVA: Effects on Social EQ and Learning IQ				
Country of upbringing × Gender				
Test Statistics				
Predictor	df	F-Statistic	p-value	Partial η^2 ⁱ
Social EQ				
Gender	1	43.31	4.72e-11	0.0008
Country of Upbringing	188	8.69	1.37e-224	0.0285
Country × Gender	131	1.53	9.34e-05	0.0036
Learning IQ				
Gender	1	17.92	2.31e-05	0.0003
Country of Upbringing	188	32.84	0.00e+00	0.0999
Country × Gender	131	1.60	1.49e-05	0.0038

ⁱ Partial $\eta^2 < 0.01$ = negligible, 0.01–0.06 = small, 0.06–0.14 = moderate, > 0.14 = large (Cohen, 1988).

Figure 5: Social EQ and Learning IQ by Gender

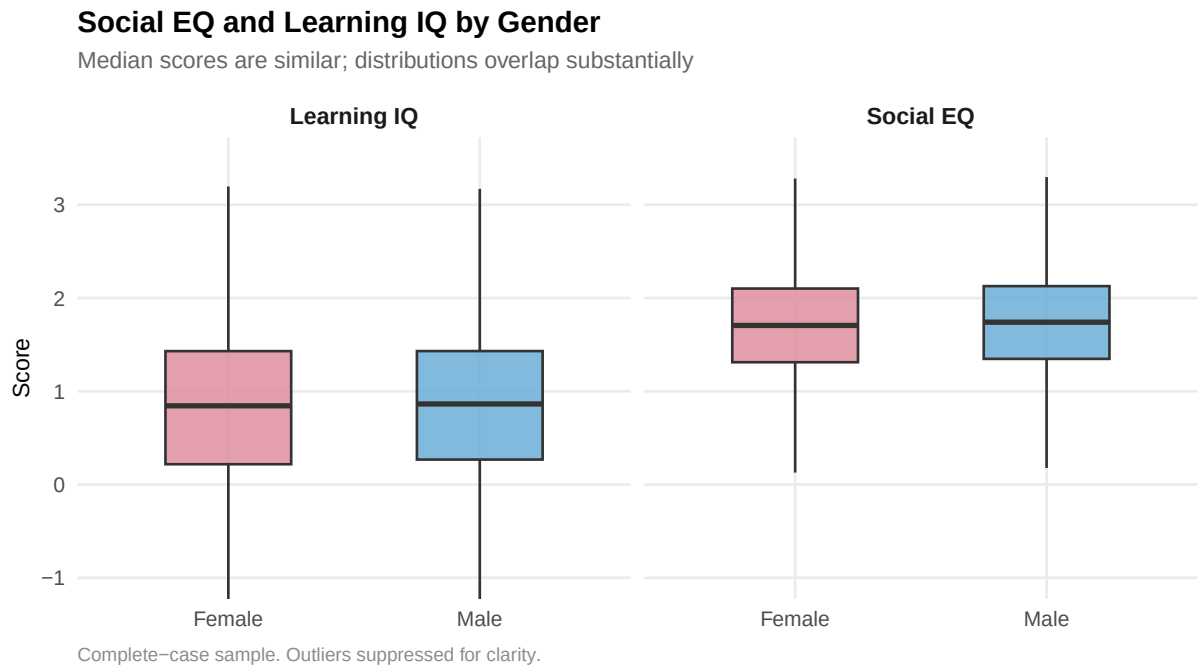


Figure 6: Social EQ and Learning IQ by Industry

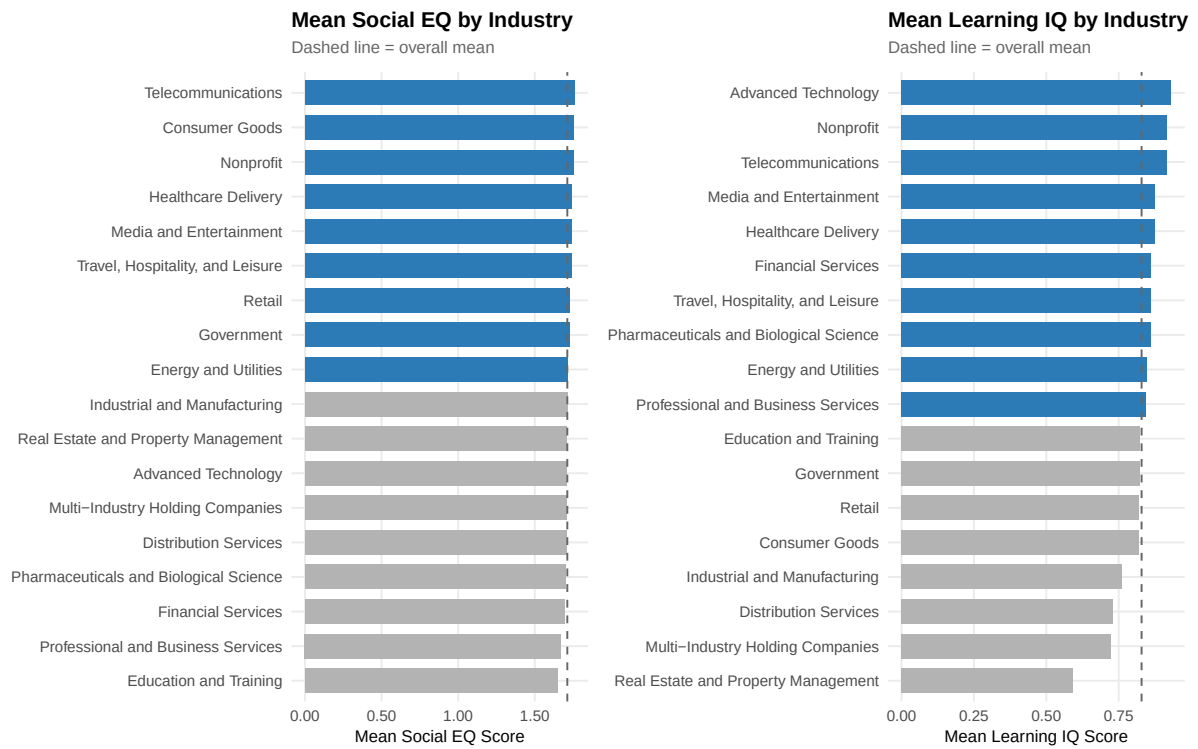
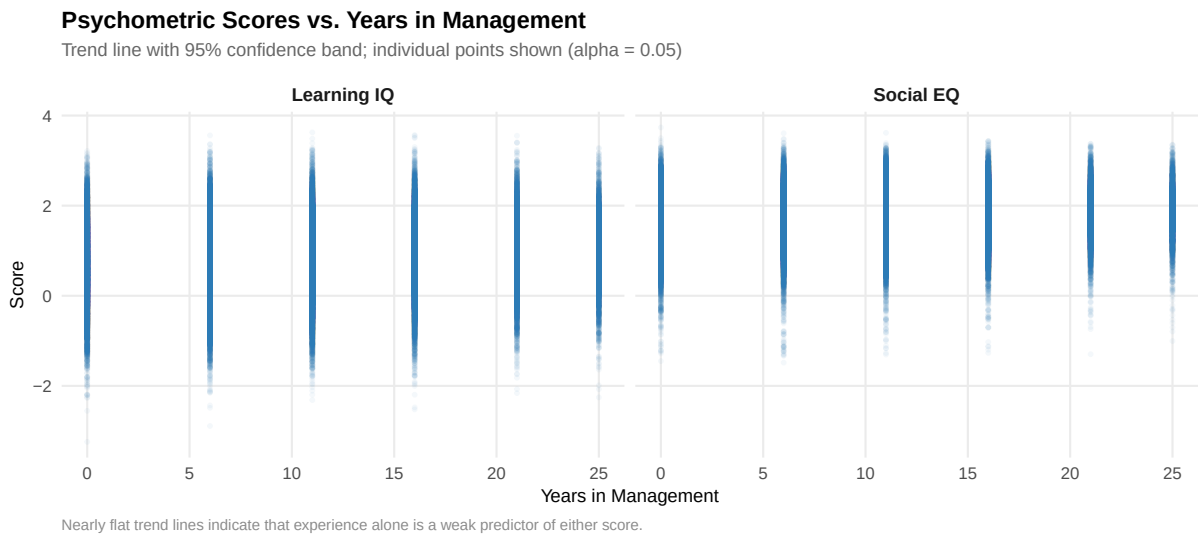


Figure 7: Psychometric Scores by Experience



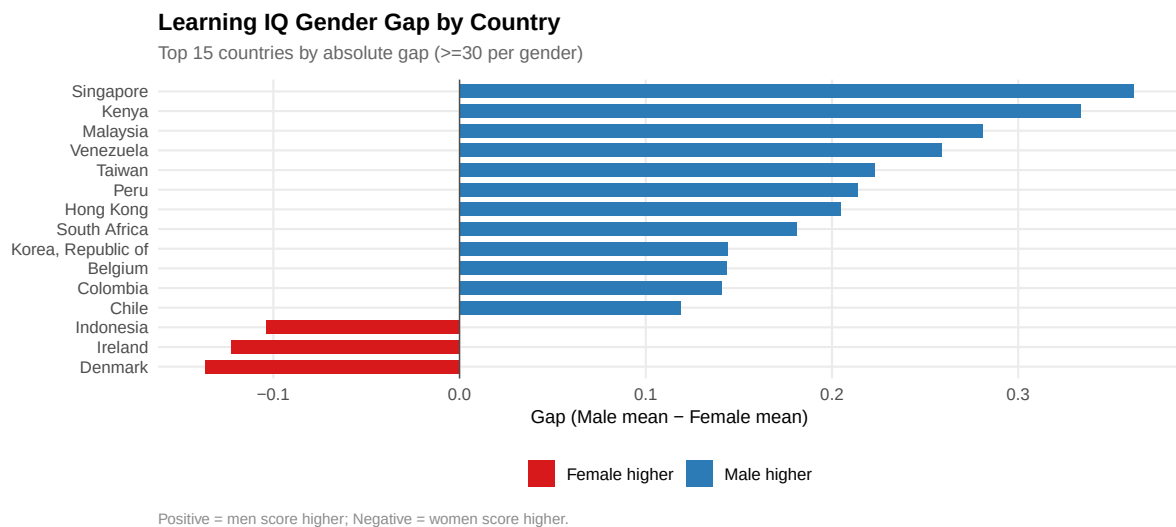
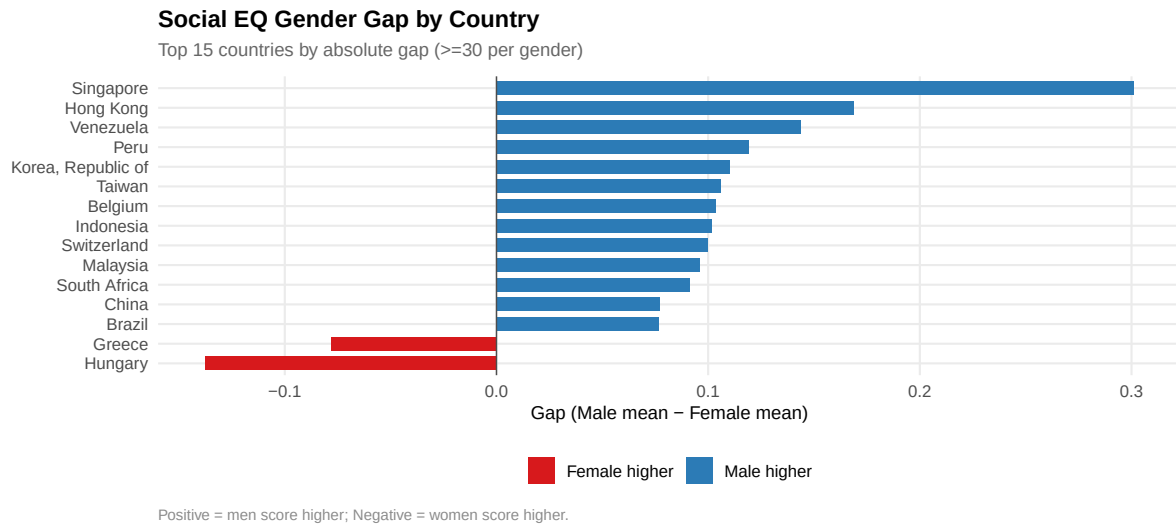


Figure 8: Social EQ and Learning IQ Gender Gaps by Country